# CODECHECK certificate 2026-029

10.5281/zenodo.23666248

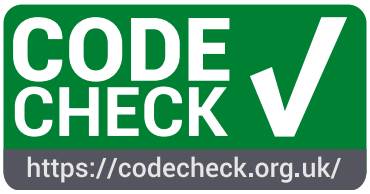



**Table 1: CODECHECK summary**

| Item | Value |
|---|---|
| Title | *Do Laypeople Endorse the Axioms of Inequality Measurement? Outcomes Report of a Preregistered Replication and Extension of Amiel and Cowell (1992) in a U.S. General-Population Sample* |
| Author(s) | Cristóbal Moya, Carsten Schröder, Philipp Lersch |
| Reference | https://doi.org/10.71240/lcyc.829503 |
| Repository | https://doi.org/10.5281/zenodo.23666248 |
| Codechecker(s) | Nataliia Petliak |
| Date of check | *2026-10-22* |
| Summary | The authors' computational reproduction workflow completed successfully and generated the expected PDF report. An initial Docker build issue was resolved by explicitly selecting the target platform. |

**Table 2: Summary of output files generated**

| File | Comment | Size (b) |
|---|---|---|
| `outcomes_report_pub.pdf` | Reproduced 25-page PDF report containing the analyses, tables, and figures. Only the first page is previewed in the certificate; the complete PDF is available in the GitHub repository. | 634352 |

This check is based on the commit `1c772ebdf0c6398878e7ce7255ff8cdd89196c7c`.

## Summary

The authors' computational reproduction workflow completed successfully and generated the expected PDF report. An initial Docker build issue was resolved by explicitly selecting the target platform.

## CODECHECKER notes

### Computational environment

The CODECHECK was performed on an Apple Silicon Mac running macOS.

The certificate-generation environment uses Python 3.12.15, Jupyter, and Typst 0.15.1.

The authors provide a Docker-based reproduction environment with R 4.6.0, Quarto 1.10.18, and an R package snapshot dated 2026-08-03 from Posit Package Manager.

Docker 28.5.1 was installed and the Docker engine was running.

### Reproduction procedure

The computational reproduction followed the four-step Docker workflow described in the authors' `README.md` .

### Steps 1–2: Download and extract the reproduction package

The reproduction materials were downloaded from the authors' OSF repository and extracted into the local CODECHECK working directory.

The downloaded archive ( `8z2uv-osfstorage-archive.zip` ) contained a nested archive ( `inequality-axioms-repro-20260831.zip` ).

The extracted package includes the Quarto source document ( `output/outcomes_report_pub.qmd` ), minimal anonymized datasets under `data/minimal/` , a `Dockerfile` , a `Makefile` , and reproduction instructions. Separate licenses are provided for the code and data.

Extraction completed successfully. No modifications were made to the authors' source files or configuration.

### Step 3: Build the Docker image

The initial Docker build was attempted from the root of the extracted reproduction package using the command specified in the README:

```
docker build -t ineq-axioms-repro:1.0 .
```

The build failed after approximately 62 seconds during the installation of Quarto 1.10.18.

The container was running on ARM64, but the Dockerfile attempted to install the AMD64 version of Quarto. This resulted in the following error:

```
dpkg: error processing archive /tmp/quarto.deb (--install):
package architecture (amd64) does not match system (arm64)
```

The Dockerfile specifies `ARG TARGETARCH=amd64` , which was not overridden during the initial build.

Consequently, the downloaded Quarto package did not match the container architecture.

The build terminated with exit code 1. The failure was caused by an architecture mismatch in the supplied Docker configuration when building on Apple Silicon.

To address the mismatch without modifying the authors' Dockerfile, the build was retried with the target platform explicitly set to AMD64:

```
docker build --platform linux/amd64 -t ineq-axioms-repro:1.0 .
```

This uses Docker's AMD64 emulation on the Apple Silicon host. The command differs from the authors' instructions only by the addition of `--platform linux/amd64`.

The second build completed successfully in 355.8 seconds, including installation of Quarto 1.10.18 and the required R packages. The resulting image was tagged `ineq-axioms-repro:1.0`.

No modifications to the authors' files were necessary.

**Step 4: Render the report**

After successfully building the Docker image for AMD64, the report was rendered using the exact command provided in the authors' README:

```
docker run --rm -v "$PWD":/work ineq-axioms-repro:1.0
```

Docker reported a platform mismatch warning because the AMD64 image was running on an ARM64 host:

```
WARNING: The requested image's platform (linux/amd64) does not match
the detected host platform (linux/arm64/v8) and no specific platform
was requested
```

The warning did not prevent execution.

Quarto completed all 89 processing steps, including the code chunks generating the analyses, tables, and figures. Pandoc and Typst then compiled the document successfully.

The process completed without errors and generated the expected output:

```
output/outcomes_report_pub.pdf
```

The authors' source files and configuration remained unchanged.

**Results and observations**

The reproduction workflow completed successfully using the authors' Docker environment with AMD64 emulation on the Apple Silicon host. The expected PDF was generated without modifying the authors' source files.

The reproduced PDF was visually inspected, and the tables and figures appeared to render correctly. A direct comparison with the published figures was not possible because they were not loading on the Lifecycle Journal website at the time of the check.

No detailed numerical comparison with the published report was performed.

## Recommendations to the authors

The Dockerfile currently downloads the AMD64 version of Quarto regardless of the container architecture, causing the default Docker build to fail on ARM64 systems such as Apple Silicon Macs.

Consider updating the Dockerfile to support multiple architectures or documenting the need to build with `--platform linux/amd64` on ARM64 hosts.

It would also be helpful to export each figure as a separate graphics file, in addition to the complete PDF report. This would make it easier to inspect individual figures, trace them back to the code that generated them, and include them explicitly in the reproduction manifest.

## Manifest files

Summary of all reproduced output files (tables, text files, figures, …).

**`output/outcomes_report_pub.pdf`**

Comment: *Reproduced 25-page PDF report containing the analyses, tables, and figures. Only the first page is previewed in the certificate; the complete PDF is available in the GitHub repository.*

# Do Laypeople Endorse the Axioms of Inequality Measurement?

## Outcomes Report of a preregistered replication and extension of Amiel and Cowell (1992) in a U.S. general-population sample

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### Abstract

The transfer principle is foundational to inequality measurement: a rank-preserving transfer of income from a richer to a poorer person must reduce inequality. Yet Amiel & Cowell (1992) found that only 35% of surveyed students endorsed it in numeric form. We test whether this finding generalizes to the U.S. adult population (H1), whether the transfer principle is the least supported of five core axioms (H2), and whether endorsing it predicts redistribution preferences (H3) or inequality aversion (H4). In a preregistered study (OSF registration 10.17605/OSF.IO/82DJS), 3,695 respondents completed the original questionnaire. Two experimental treatments—proportional visual aids and a performance bonus—tested whether low measured support is an artifact of presentation or effort. A robustness sample ($N$ = 2,125) applies the registered attention-check and speeding exclusions. Tests comprised one-tailed proportion tests (H1), a logistic mixed model over axioms with respondent random intercepts (H2), and regressions of redistribution preferences, tolerated efficiency losses in a leaky-bucket task, and incentivized donation on transfer-principle support (H3, H4).

Numeric transfer-principle support was 26.3% (95% CI [24.9%, 27.8%]) and verbal support 30.0%, both below H1′s 66% threshold; all four other axioms drew significantly more support (numeric-format support 32.1%–55.9%), so H1 and H2 were supported. Transfer-principle support did not predict public and private redistribution preferences, tolerated efficiency losses, or donation of an earned bonus; H3 and H4 were not supported. In the case of redistribution preferences and efficiency-loss outcomes, the confidence intervals exclude positive differences larger than 0.26 scale points (1–7) and 0.05 (0–1 index). The same pattern holds in the robustness sample. Low support for the transfer principle generalizes beyond student samples, but endorsement predicts neither redistributive preferences nor behavior.

**Keywords:** income inequality; transfer principle; inequality axioms; redistribution preferences; inequality aversion; leaky bucket; replication.

### 1 Introduction

Virtually every applied measure of income inequality is built on a small set of axioms. Chief among them is the transfer principle: a rank-preserving transfer of income from a richer to a poorer person must reduce inequality. Also known as the Pigou–Dalton principle (Pigou, 1912), it is what makes the Lorenz ordering, the Gini coefficient, and the family of relative inequality indices behave the way analysts expect (Clark & D'Ambrosio, 2015).

Amiel and Cowell (1992) subjected this and four other core axioms (scale invariance, translation invariance, the population or replication principle, and decomposability) to an empirical test outside the expert community. They asked university students in the United States, Germany, the United Kingdom, and Israel to compare pairs of income distributions, once in numerical and once in verbal form. Their results were striking: only 35% endorsed the transfer principle in the numerical comparison, and roughly 60% in the verbal statement. Support for the other axioms was similarly limited. If the axioms that experts treat as self-evident are endorsed by only a minority of respondents, there is a substantive gap between the foundations of inequality measurement and the public understanding of inequality.

We ask whether that divergence between expert axioms and lay perceptions holds in a general adult population, and whether endorsing the transfer principle carries any broader distributive meaning in the public's understanding of inequality. Four considerations motivate these questions. First, the original

## Citing this document

Nataliia Petliak (*2026-10-22*). CODECHECK Certificate 2026-029. CODECHECK Community on Zenodo. 10.5281/zenodo.23666248

## About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely availableby following the links in this document, the reader can then study for themselves the code and data.

## About this document

This document was created using codecheck-py (a Python-based template for creating CODECHECK certificates). The CODECHECK details are filled into a jupyter notebook which is then converted into Markdown via nbconvert. Afterwards it gets converted into Typst using cmarker and then into PDF using Typst. `sh notebook_to_pdf.sh` will regenerate the report file.

```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

```
Pygments  2.21.0
debugpy 1.8.22
idna  3.20
pyzmq 27.2.0
packaging 26.3
platformdirs  4.13.5
urllib3 2.8.0
psutil  7.2.2
jupyter_core  5.9.1
ipykernel 7.4.0
appnope 1.0.0
tornado 6.6
typing_extensions 4.16.0
session-info2 0.4.2
comm  0.2.3
charset-normalizer  3.5.2
six 1.17.0
python-dateutil 3.0.0
pandas  3.0.6
certifi 2026.10.19
jupyter_client  8.10.0
ipython 9.17.1
PyYAML  6.0.3
numpy 2.5.4
requests  2.34.2
traitlets 5.16.1
----  ----
Python  3.12.15 (main, Sep 30 2026, 22:04:19) [Clang 21.0.0 (clang-2100.3.34.2)]
OS  macOS-26.6.2-arm64-arm-64bit
CPU 14/14 logical CPU cores, arm
GPU No GPU found
Updated 2026-11-02 14:30
```

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